

SECTION A

(25 marks)

1. Solve the following:

(a) $\lim_{t \rightarrow 1} \frac{t^3 + t^2 - 5t + 3}{t^3 - 3t + 2}$ (4 marks)

(b) $\lim_{x \rightarrow 0} \frac{x}{\sqrt{1+x+x^2}-1}$ (5 marks)

2. (a) By using the definition of $f'(x) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$, find the values of a and b such that $f'(1)$ exist if

$$f(x) = \begin{cases} |x-1| & , \quad x < 1 \\ ax+b & , \quad x \geq 1 \end{cases} \quad (6 \text{ marks})$$

- (b) Find the derivative of the following functions

$$y = \frac{\sin x}{1 - \cos x} \quad (4 \text{ marks})$$

3. A closed right circular cylindrical container of radius r and height h is to be constructed with volume $4,000\text{cm}^3$. The cost for the construction is RM1.00 per cm^2 for the curved surface while RM2.00 per cm^2 for the top and bottom surfaces.

State h in terms of r and hence, find the radius of the cylinder so that the cost of the construction is minimum. (6 marks)

SECTION B

(75 marks)

1. Given the complex number $z = \frac{4}{i - \sqrt{3}}$. Find the modulus and argument of z . Express z in the form $r(\cos \theta + i \sin \theta)$. (5 marks)

2. (a) Solve $\log_p(5 + x^2) = \log_{\sqrt{p}}(3 - x)$. (6 marks)

- (b) Find the solution set of the inequality $\frac{x+8}{3-x} \leq 4$. (6 marks)

3. Given $f(x) = \sin(2x - \pi)$ for $0 \leq x \leq 2\pi$.

- (a) Sketch the graph of $f(x)$. (3 marks)

- (b) Find the domain and range of $f(x)$. (2 marks)

4. Function f and g are defined as follow:

$$f(x) = x^2 - 8x + 15, \quad x \leq 4$$

$$g(x) = 4 + \ln x, \quad x \geq 0$$

- (a) Find $f^{-1}(x)$, $g^{-1}(x)$ and their domain. (7 marks)

- (b) Hence, find $(f \circ g)^{-1}(x)$. (3 marks)

- (c) Find $h(x)$ where $(h \circ g)(x) = \ln x^2 + 1$. (4 marks)

5. Solve the following:

(a) Determine the continuity of $f(x)$ at $x=1$.

$$f(x) = \begin{cases} \frac{x^2 + x - 2}{x - 1} & , x > 1 \\ 1 & , x = 1 \\ x^3 + 2 & , x < 1 \end{cases}$$

(5 marks)

(b) Find the vertical and horizontal asymptotes for graph of $f(x) = \frac{x^2 + 1}{x - 1}$. Hence sketch the graph of $f(x)$. (9 marks)

6. The curve with equation $y = \frac{a + 4x - x^2}{2 - ax - x^2}$, where a is constant, passes through the point (1,1).

(a) Calculate the value of a . (2 marks)

(b) Hence, find all the asymptotes of the curve. (4 marks)

(c) Express y in partial fractions. Hence, find the values of $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ when $x = 0$.

(8 marks)

7. (a) Find the local minimum and the local maximum of the curve $f(x) = x + \frac{9}{x}$. (5 marks)

(b) Water is poured into vase at a constant rate of $80\pi\text{cm}^3\text{s}^{-1}$. When the depth of water is h cm, the volume of water, $V\text{cm}^3$ is given by $V = 4\pi h(h + 4)$. Find the rate of change of the depth of water in the vase when $h = 6\text{cm}$. (5 marks)

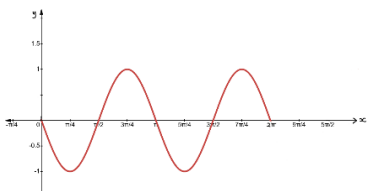
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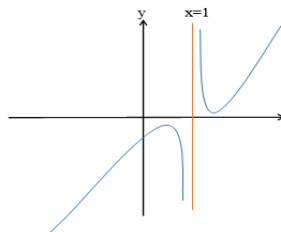
ANSWER

SECTION A

1. (a) $\frac{4}{3}$ (b) 2
2. (a) $a = -1, b = 1$ (b) $\frac{-1}{1 - \cos x}$
3. $h = \frac{4000}{\pi r^2}, r = 6.83$

SECTION B

1. $|z| = 2, \text{Arg}(z) = -2.618 \text{ rad or } -\frac{5}{6}\pi \text{ rad } z = 2[\cos(-2.618) + i\sin(-2.618)]$
2. (a) $x = \frac{2}{3}$ (b) $\left\{x : x \leq \frac{4}{5} \text{ or } x > 3\right\}$
3. (a)  (b) $D_f = [0, 2\pi], R_f = [-1, 1]$
4. (a) $f^{-1}(x) = 4 - \sqrt{x+1}, D_{f^{-1}} = [-1, \infty]$ (b) $e^{-\sqrt{x+1}}$ (c) $h(x) = 2x - 7$
 $g^{-1}(x) = e^{x-4}, D_{g^{-1}} = \mathbb{R}$
5. (a) $f(x)$ discontinuous at $x = 1$ (b) The vertical asymptote is $x = 1$
 No horizontal asymptote for $f(x)$



6. (a) $a = -1$ (b) V.A = 2, -1, H.A $y = 1$
- (c) $\frac{-1+4x-x^2}{2+x-x^2} = 1 - \frac{2}{1+x} + \frac{1}{2-x}, \frac{dy}{dx} = \frac{9}{4}, \frac{d^2y}{dx^2} = \frac{-15}{4}$
7. (a) (3,6) minimum, (-3,-6) maximum (b) $\frac{5}{4} \text{ cms}^{-1}$

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